

Return-Decay Residuals and Tail-Risk Forecasting: Timing Artifacts, Conditional Inference, and Cross-Market Evidence

Active revision | Summary: 27 September 2026

Does a return-decay residual contain useful future tail-risk information, or can timing and model selection manufacture the appearance of predictability?

CENTRAL IDEA

Separate a proxy's algebra, its information timing, and its forecasting evaluation. A residual formed as fitted decay minus Sharpe is a reparameterization when both components are already observed; regularization can nevertheless change fitted predictions.

MAIN FINDING

A Gaussian construction produces a same-month crash-risk ratio of 1.61 without future predictability. In the frozen regional forecast comparison, pooled log loss worsens by 0.00704 with frozen tuning and 0.00233 with annual reselection; other metrics and matched-history comparisons need not agree.

EVIDENCE AND SCOPE

The revision audits US factors and six factors across Europe, Japan and Asia Pacific excluding Japan, including a 2025-August 2026 extension. It separates history effects from hyperparameter reselection and examines sparse-event validation and influential factor cases.

LIMITS

All 48 reported conditional intervals include zero but still allow nontrivial gains. This does not prove equivalence or universal proxy failure. ETF activity is not direct ownership-based identification of crowding, and contemporaneous association is not prospective forecasting evidence.

TAKEAWAY

A financial proxy needs a time-consistent forecast comparison and an explicit information baseline before its apparent signal can be interpreted.

한국어 핵심 요약

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